

# Plaksha Dynamics and Simulation

## Mid-term Exam

Saturday March 1, 2025 10 AM - 1 PM

No calculators, books or notes allowed.

3 graded Problems, 90 minutes (+ 90 minutes over time)

### \*\*\* How to get the highest score? \*\*\*

Please do these things:

- **Your written work.** Start each problem on a clean sheet.  
— Put **\*your name and \*problem number and \*page number** on the top of every side of every sheet.
- Draw **Free body diagrams** whenever force, moment, linear momentum, or angular momentum balance are used.
- Use correct **vector notation**.
- A+ Be (I) neat, (II) clear and (III) well organized.
- TIDILY REDUCE and box in your answers (Don't leave simplifiable algebraic expressions).
- >> Make appropriate `Matlab` code clear and correct.  
You can use shortcut notation like " $\phi_7 = 2\pi$ " instead of, say, "`phi(7) = 2*pi;`".  
Small syntax errors will have small penalties.
- ↗ Clearly **define** any needed dimensions ( $\ell, h, d, \dots$ ), coordinates ( $x, y, r, \theta, \dots$ ), variables ( $v, m, t, \dots$ ), base vectors ( $\hat{i}, \hat{j}, \hat{e}_r, \hat{e}_\theta, \hat{\lambda}, \hat{n}, \dots$ ) and signs ( $\pm$ ) with sketches, equations or words.
- **Justify** your results so I can distinguish an informed answer from a guess. If you quote a fact that a grader might doubt your understanding of, explain it. Especially if it is not commonly used.
- 👁 If a problem seems *poorly defined*, clearly state any reasonable assumptions (that do not oversimplify the problem).
- ≈ Work for **partial credit** (from 60–100%, depending on the problem)
  - Put your answer is in terms of well defined variables even if you have not substituted in the numerical values.
  - Reduce the problem to a clearly defined set of equations to solve.
  - Provide computer code, or pseudo-code which would generate the desired answer, and explain the nature of the output (unless specifically precluded).

Problem 1: \_\_\_\_/50

Problem 2: \_\_\_\_/50

Problem 3: \_\_\_\_/50

Problem 4: \_\_\_\_/50

— Do any three. If you do more, your exam score will be based on your 3 best answers. —

**1) Dashpot in 3D.** Two masses  $m_A$  &  $m_B$  are at positions  $\vec{r}_A$  &  $\vec{r}_B$  and have velocities  $\vec{v}_A$  &  $\vec{v}_B$ . They are connected by a linear dashpot that has negligible mass and has dashpot constant  $c$ . The only relevant forces in this problem are from the dashpot. In terms of some or all of the quantities above, find the acceleration  $\vec{a}_A$  of mass A. You need not explicitly find any vector components.

**2) Mass and zero rest-length spring. 2D** Given that a mass  $m$  hangs from a zero-rest-length spring  $k$  with one end attached to the origin. It is released from rest at  $\vec{r} = \vec{r}_0$ . Gravity  $-g\hat{j}$  also acts on the mass. In terms of some or all of the quantities above, find  $y(t)$ .

**3) Four planets move in a circle. 2D.** Four equal masses  $m$  are mutually attracted by inverse square gravity with gravitational constant  $G$ . There are no other forces. Assume they initially form a square, with each of them is on the  $+x$ ,  $-x$ ,  $+y$  or  $-y$  axis, a given distance  $d$  from the origin (That is, the sides of the square are length  $\sqrt{2}d$ ). What are the initial velocities of the four masses so that, when the governing equations are solved, the particles will all move on a circle with radius  $d$ ? (Please draw one or more clear sketches which you can use to describe your reasoning and your answer.)

**4) Angular momentum.**

a) For a collection of particles, show that the total angular momentum is the some of two terms: One concerning the motion of the center of mass and one concerning motion relative to the center of mass. Define the rates of changes of the quantities above as  $\dot{\vec{H}}_G$  and  $\dot{\vec{H}}_{/G}$ .

b) Consider these three quantities

i)  $\sum \vec{r}_i \times \vec{F}_i$

ii)  $\vec{r}_G \times \sum \vec{F}_i$

iii)  $\sum \vec{r}_{i/G} \times \vec{F}_{i/G}$ .

How many correct equations can you write that involve equating these three terms, or their sums, with the two terms in part (a) or their sums? Zero credit for unjustified guesses.